Chapter 3 – First Normal Form (1NF)

Chapter Objectives

After completing this chapter, you will understand:

- What is First Normal Form (1NF)?
- Why was 1NF introduced?
- What is Atomicity?
- What are Repeating Groups?
- Why are Multi-valued Attributes bad?
- How to convert a table into 1NF?
- Primary Key in 1NF
- Composite Key in 1NF
- Real-world examples
- MySQL implementation
- Common mistakes
- Interview questions
- Practice questions

Before Learning 1NF

Let's understand **why 1NF exists**.

Imagine you are designing a database for a college.

You create this table.

Student Table

StudentID	Name	Phone Numbers
101	Akshat	9876543210, 9999999999
102	Rahul	8888888888

Looks fine?

Actually...

This table has a serious problem.

The Problem

Look carefully.

```
Phone Numbers

9876543210,9999999999
```

How many values are stored inside one cell?

Two

One database cell should store

One value

Not multiple values.

Why Is This a Problem?

Suppose the principal asks

> "Find students having phone number 9999999999."

Your SQL query becomes difficult.

You have to search inside a comma-separated string.

Instead of

```
=
```

you need string functions like

```
LIKE '%9999999999%'
```

Slow.

Difficult.

Error-prone.

Another Problem

Suppose Akshat gets one more phone number.

Now

```
9876543210

9999999999

8888888888
```

All stored in one cell.

The cell keeps growing.

Databases are **not designed** for this.

Definition of 1NF

A table is in **First Normal Form (1NF)** if:

1. Every column contains **atomic (single) values**.
2. There are **no repeating groups**.
3. Each row can be uniquely identified by a key.

What is Atomicity?

This is the **most important concept** in 1NF.

Definition

Atomic means:

> **Indivisible**
>
> **Cannot be broken into smaller meaningful values within the context of that column.**

Real-Life Meaning

Imagine buying chocolates.

One chocolate

↓

Atomic

A box containing 10 chocolates

↓

Not atomic

Because the box contains multiple chocolates.

Database Meaning

One database cell

↓

One value

Only.

Examples

Correct (Atomic)

StudentID	Name
101	Akshat

Each cell contains exactly one value.

Wrong

StudentID	Subjects
101	SQL, Python, Java

One cell

Three values

Not Atomic.

More Examples

Wrong

Employee	Skills
Akshat	Java, Python, React

Correct

Employee	Skill
Akshat	Java
Akshat	Python
Akshat	React

Each row stores one skill.

What is a Repeating Group?

Suppose we design a table like this.

StudentID	Phone1	Phone2	Phone3
101	9876543210	9999999999	8888888888

Notice

Phone1

Phone2

Phone3

Same type of information is repeated.

This is called a

Repeating Group

Why Repeating Groups Are Bad

Suppose one student has

5 phone numbers.

Need

Phone5 column.

Suppose another student has

10 phone numbers.

Need

Phone10 column.

The table keeps changing.

Bad database design.

Better Design

Instead

Create another table.

Student

StudentID	Name
101	Akshat

StudentPhone

StudentID	Phone
101	9876543210
101	9999999999
101	8888888888

No repeating columns.

Unlimited phone numbers.

Better design.

Multi-Valued Attributes

A column containing more than one value is called a

Multi-Valued Attribute

Example

```
Skills

Java, Python, SQL
```

Wrong.

Correct

Employee	Skill
Akshat	Java
Akshat	Python
Akshat	SQL

How to Convert to 1NF

Example

Before

OrderID	Products
1	Laptop, Mouse, Keyboard

Split the values.

After

OrderID	Product
1	Laptop
1	Mouse
1	Keyboard

Now

One value

Per cell.

But Wait...

You asked an excellent question earlier.

Look here.

StudentID	Name	Phone
101	Akshat	9876543210
101	Akshat	9999999999

StudentID repeats.

Is that allowed?

Yes.

Why?

1NF **does not say** every value must be unique.

It says

Every **row** must be uniquely identifiable.

How?

By choosing the correct key.

Example

Table

StudentID	Phone
101	9876543210
101	9999999999

Can StudentID alone be Primary Key?

No.

Because

101 repeats.

Instead

Primary Key

```
(StudentID, Phone)
```

Check

(101,9876543210)

↓

Unique

(101,9999999999)

↓

Unique

So

Composite Primary Key

Works perfectly.

Important Interview Point

1NF does NOT require every column to be unique.

It only requires

The rows can be uniquely identified.

SQL Example

Wrong Design

```
CREATE TABLE Student(
StudentID INT,
Name VARCHAR(50),
Phones VARCHAR(100)
);
```

Data

```
101

Akshat

9876543210,9999999999
```

Not 1NF.

Better Design

```
CREATE TABLE Student(
StudentID INT PRIMARY KEY,
Name VARCHAR(50)
);
```

```
CREATE TABLE StudentPhone(
StudentID INT,
Phone VARCHAR(15),
PRIMARY KEY(StudentID,Phone),
FOREIGN KEY(StudentID)
REFERENCES Student(StudentID)
);
```

Why Is This Better?

Now

One phone

One row.

Easy searching.

Easy updating.

Easy deleting.

Real-Life Example

Amazon

Wrong

OrderID	Products
101	Laptop, Mouse, Keyboard

Correct

OrderID	Product
101	Laptop
101	Mouse
101	Keyboard

Hospital Example

Wrong

Patient	Diseases
Akshat	Fever, Cold

Correct

Patient	Disease
Akshat	Fever
Akshat	Cold

Common Mistakes

Mistake 1

Thinking

Repeated rows

Mean

Not 1NF.

Wrong.

Repeated values are allowed.

Only duplicate **entire rows** are problematic, and more importantly, every row should be uniquely identifiable using a key.

Mistake 2

Thinking

StudentID

Must always be Primary Key.

Wrong.

Sometimes

Composite Key

is needed.

Mistake 3

Thinking

Comma-separated values

Are acceptable.

Wrong.

Never store

`Java,Python,SQL`

inside one column.

How to Check if a Table Is in 1NF

Ask these questions.

Question 1

Does every cell contain only one value?

Yes

↓

Continue

No

↓

Not 1NF.

Question 2

Are there repeating columns like

Phone1

Phone2

Phone3

Yes

↓

Not 1NF.

Question 3

Can every row be uniquely identified?

Yes

↓

1NF.

No

↓

Fix the key.

Interview Questions

Q1

What is 1NF?

Answer

A table is in First Normal Form if every attribute contains atomic values, there are no repeating groups, and each row can be uniquely identified.

Q2

What is Atomicity?

Answer

Atomicity means one cell contains one indivisible value.

Q3

Can values repeat in 1NF?

Answer

Yes.

Values like names or StudentIDs (when they are not the sole primary key in that table) may repeat.

What matters is that the **row can be uniquely identified by a key**.

Q4

Can StudentID repeat?

Yes.

If

Primary Key

`(StudentID,Phone)`

then StudentID can repeat.

Q5

Can a table with a single primary key be in 1NF?

Yes.

Absolutely.

Practice Questions

Question 1

Is this in 1NF?

Student	Subjects
Akshat	SQL,Python

Answer

No.

Question 2

Convert it to 1NF.

Answer

Student	Subject
Akshat	SQL
Akshat	Python

Question 3

Is this in 1NF?

Student	Phone1	Phone2

No.

Repeating Groups.

Question 4

Which is better?

`Phone1`

`Phone2`

`Phone3`

OR

`StudentPhone Table`

Answer

StudentPhone Table.

Chapter Summary

1NF Rules

- One cell = One value (Atomicity).
- No repeating groups (e.g., Phone1, Phone2, Phone3).
- No multi-valued attributes (e.g., "Java, Python").
- Every row must be uniquely identifiable by a key.

Memory Trick

Think of **1NF** as a **clean spreadsheet**.

Ask four questions:

1. **One cell = One value?** ✓
2. **Any comma-separated values?** ✗

3. **Any repeating columns like Phone1, Phone2? ✗**

4. **Can I uniquely identify every row with a key? ✓**